

ASEN 5090
Fall 2026
HW 3
Jash Bhalavat

Problem 1a

This section does not ask for any results and only asks to extract the data. Here is a snippet of the code to show how the GPS data was extracted from the provided rinex file:

%% Problem 1a

```
% Set filename and use rinexread to extract GPS data
nist_rinex_filename = 'NIST00USA_R_20262310000_01D_30S_M0.rnx';
rinex_data = rinexread(nist_rinex_filename).GPS;

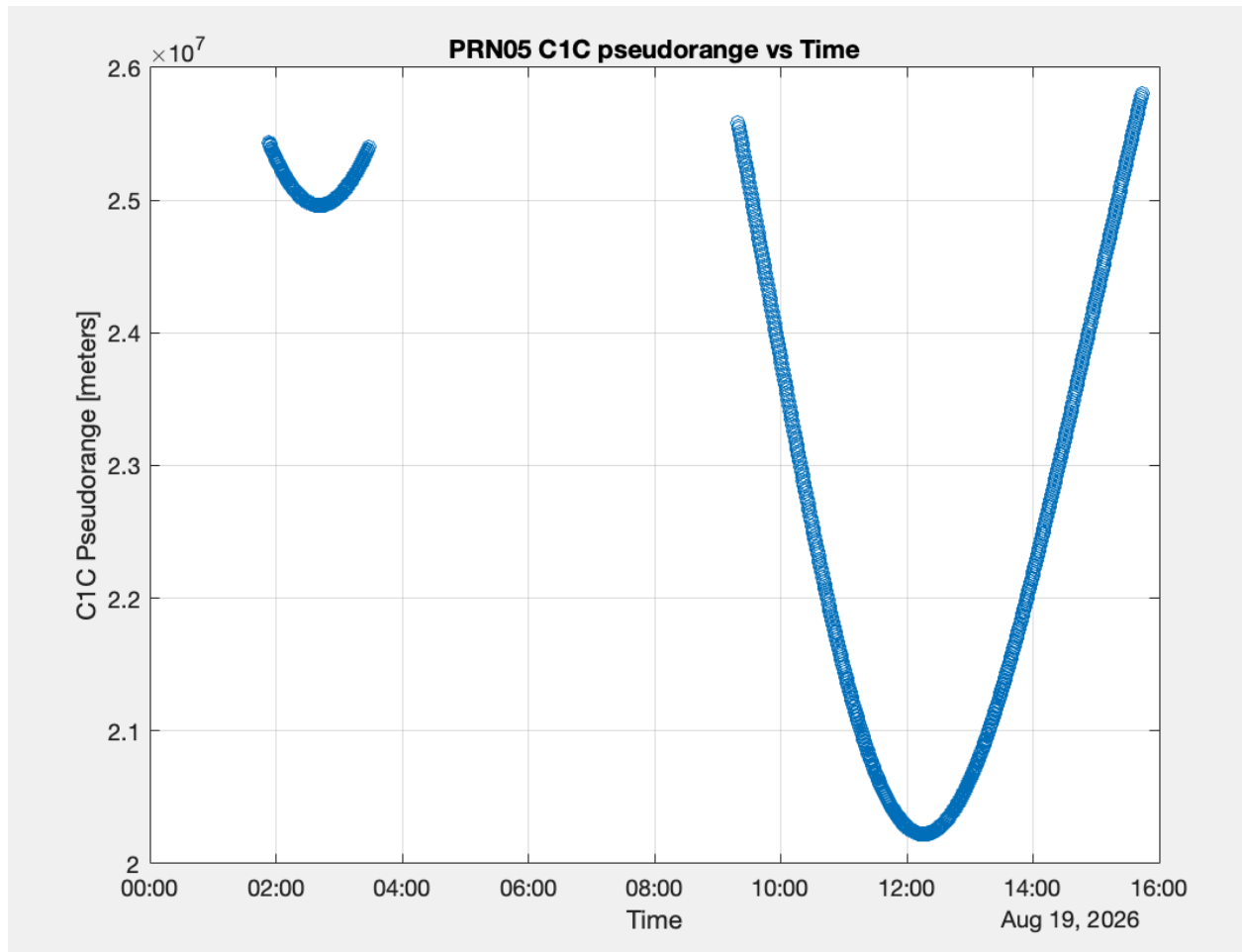
% Get time and convert data to arrays
time = rinex_data.Time;
rinex_data_array = table2array(rinex_data);

% Only extract PRN05 data and time
prn05_data = rinex_data_array(rinex_data_array(:,1) == 5, :);
prn05_time = time(rinex_data_array(:,1) == 5);
```

Matlab's rinexread() function was used to extract the GPS measurements.

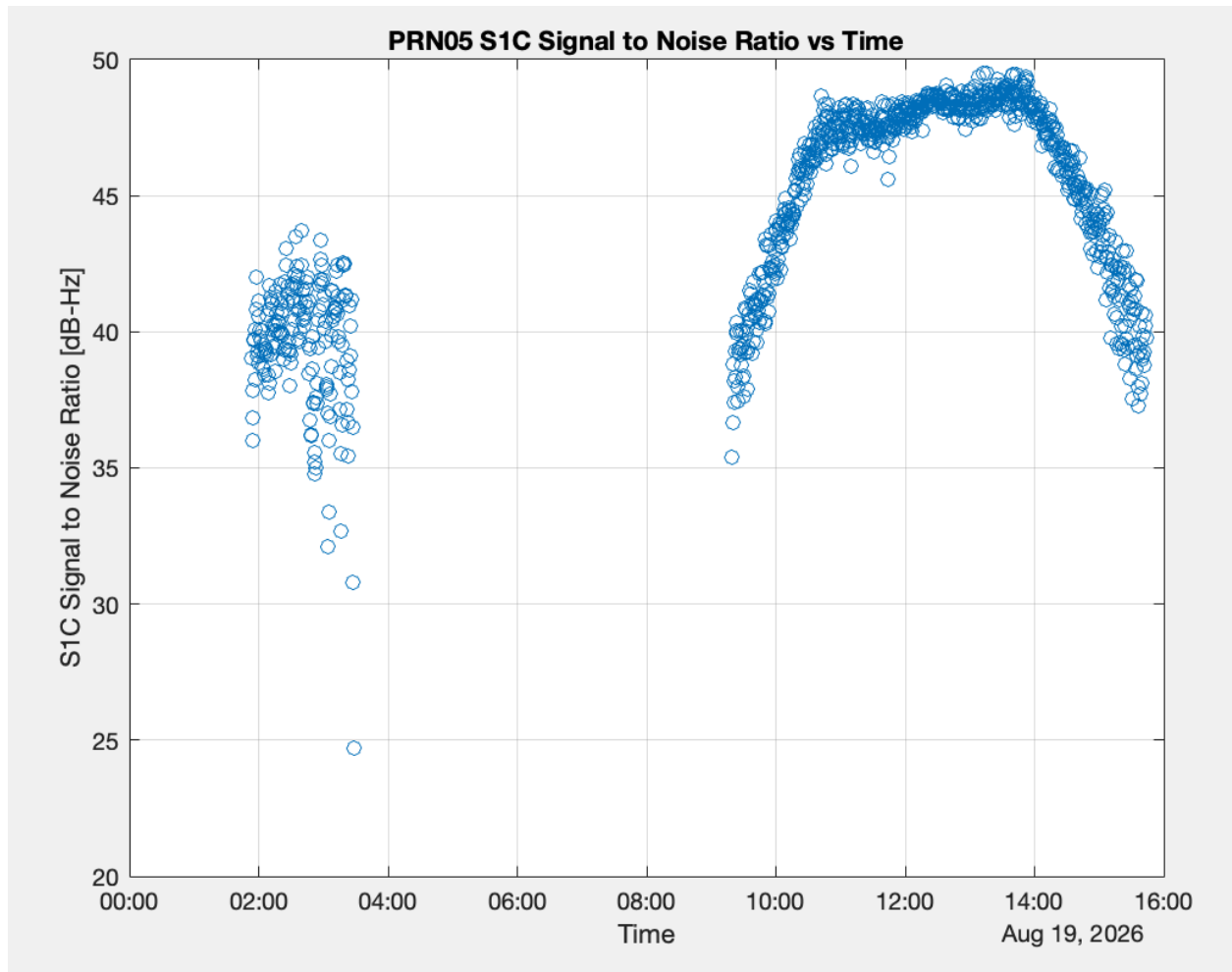
Problem 1b

Here's the C1C Pseudorange vs time for PRN05:



Problem 1c

Here's is the plot for the signal to noise ratio S1C for PRN05 vs time:



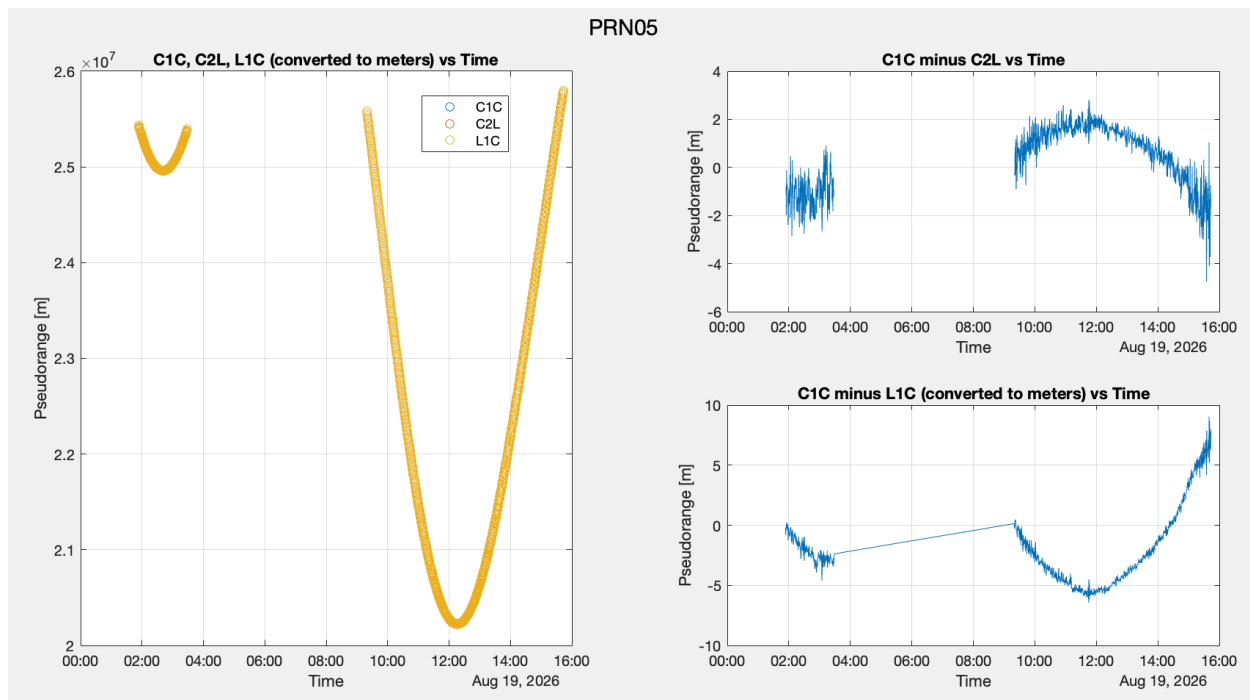
Problem 1d

Converting carrier phase L1C (C/A code derived L1 carrier phase derived from C channel) to meters:

Pseudorange = L1C (cycles) x Wavelength

Wavelength is calculated by c/f where c is speed of light in vacuum and f is frequency of L1 signal (1575.42 MHz).

Additionally, C2L (L2C pseudorange derived from the L channel) is also plotted in the plot below:



As seen on the plots on the right, the difference between C1C and C2L and L1C have been added. Those differences are in the order of meters. It can be seen that L1C error is slightly higher than C2L which makes sense because L1C is not inherently a pseudorange measurement. It is converted from the carrier phase measurement.

Problem 2a

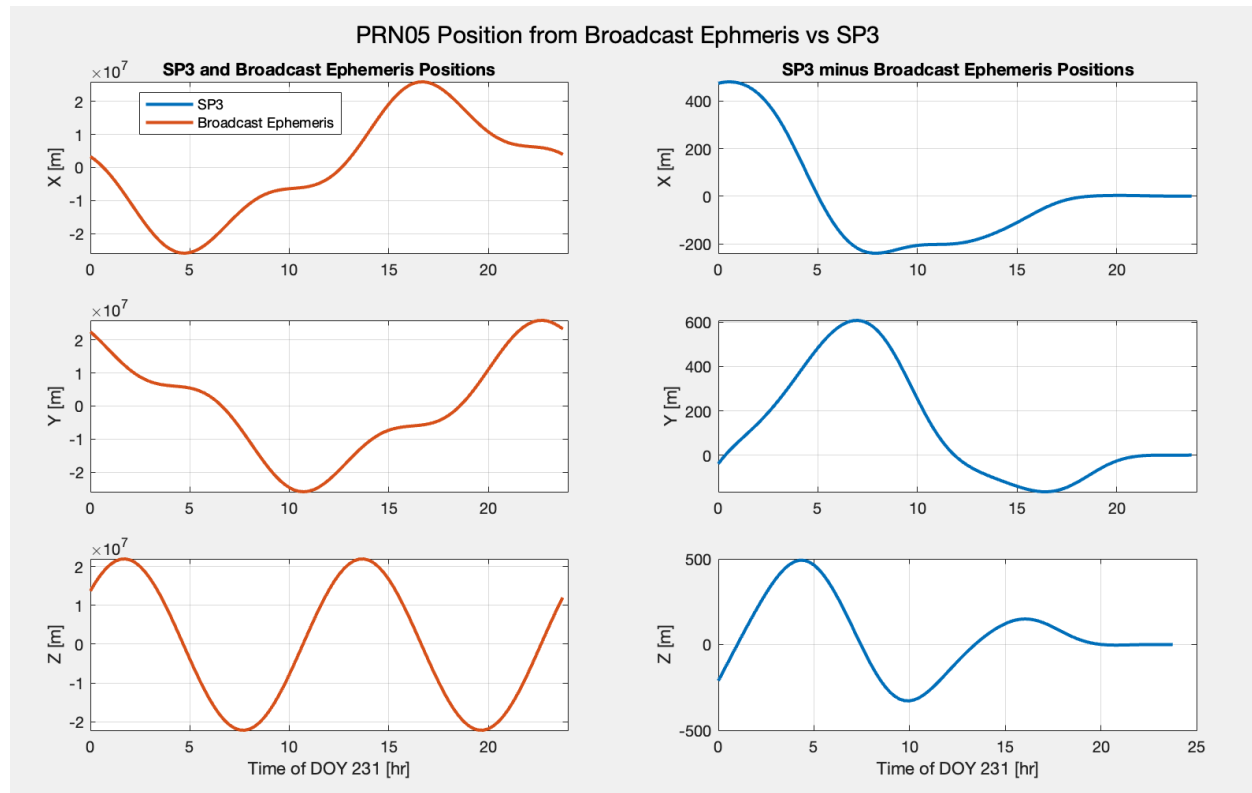
This section does not ask for any results and only asks to extract the data. Here is a snippet of the code to show how the broadcast data was converted to a numerical array called “ephem_data”:

%% Problem 2a

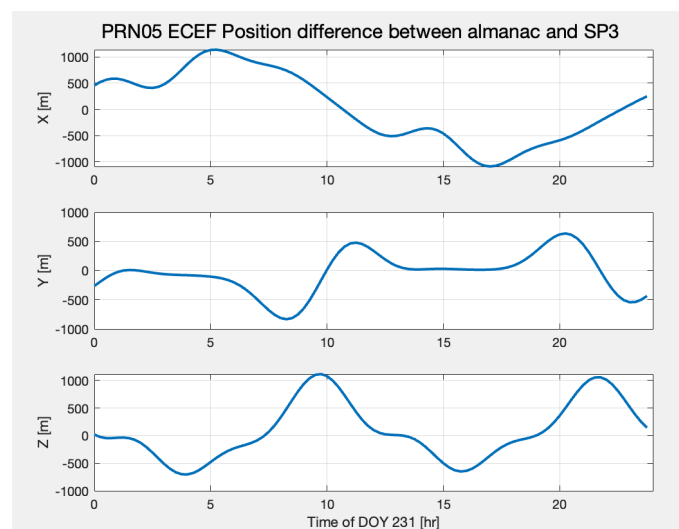
```
% Read broadcast file and save data into array|
broadcast_ephem_filename = 'brdc2310.26n';
ephem_data = read_clean_GPSbroadcast(broadcast_ephem_filename);
```

Problem 2b

Here is the position of PRN05 based on the ephemeris data for the times given in the SP3 file.



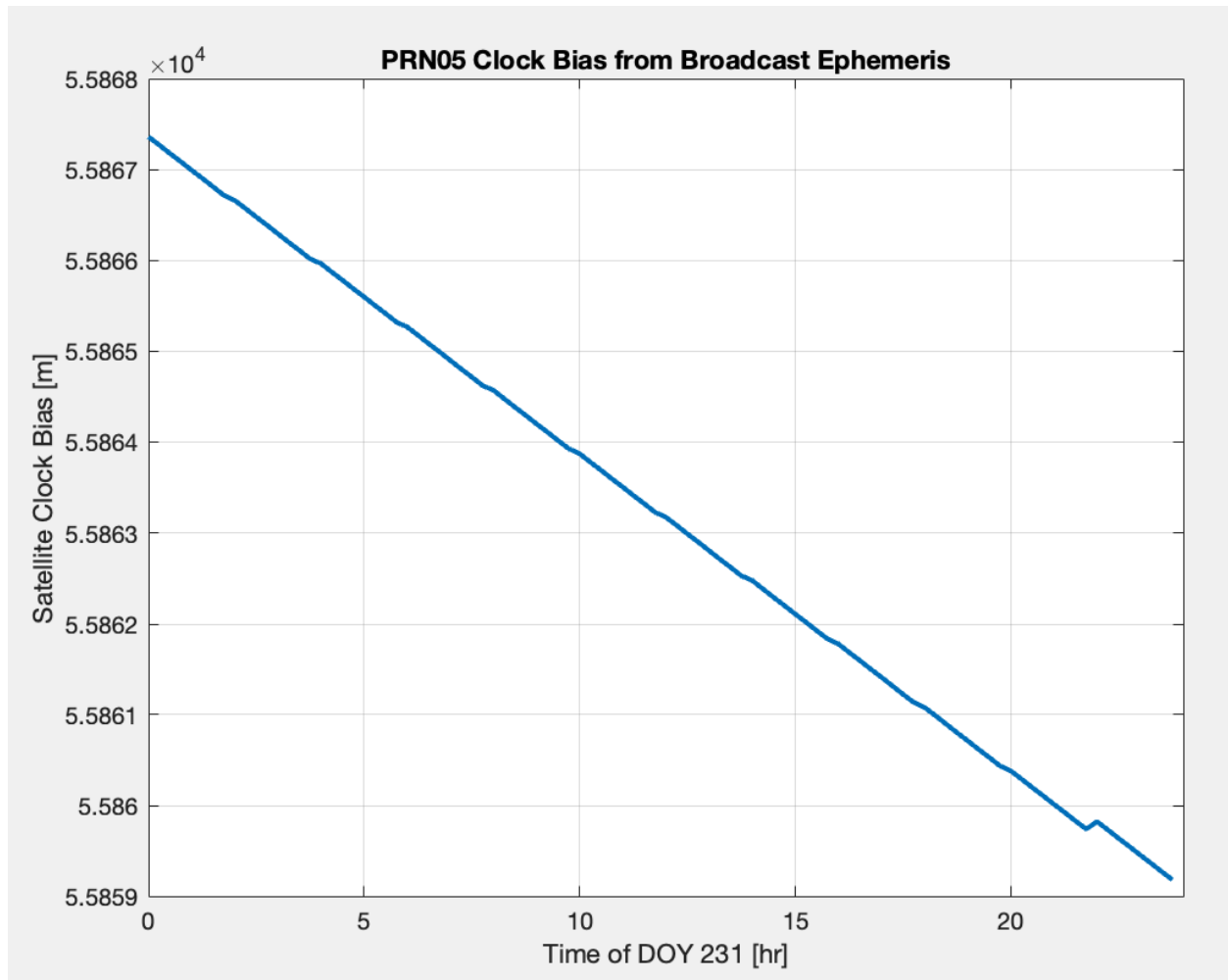
The difference between SP3 (truth) and ephemeris derived positions is in the order of 100s of meters. From HW2, here is the difference between SP3 and almanac:



The difference above is around 1 km, so broadcast data is slightly better than almanac.

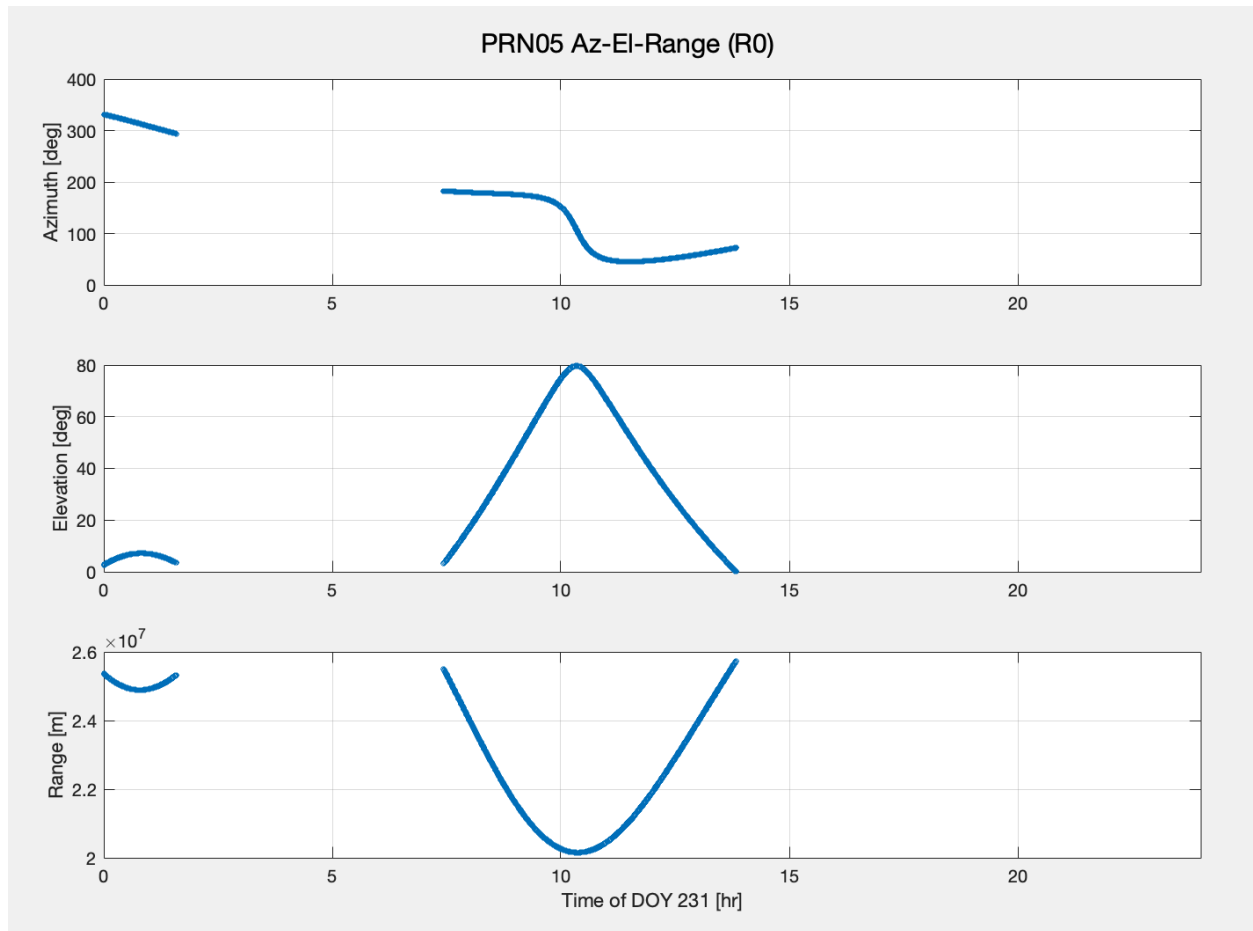
Problem 2c

Here's the satellite clock bias based on the a0 and a1 parameters vs time:



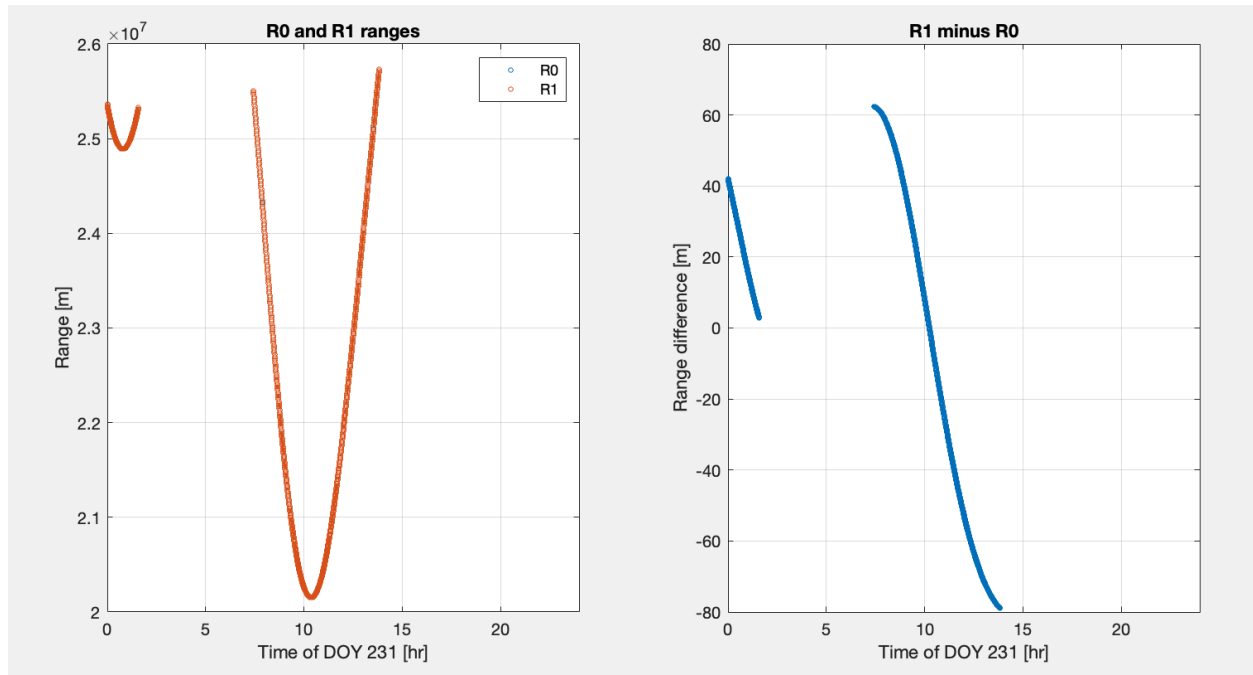
Problem 3a

Here's the Azimuth, Elevation, Range (R0) for PRN05 from NIST:



Problem 3b

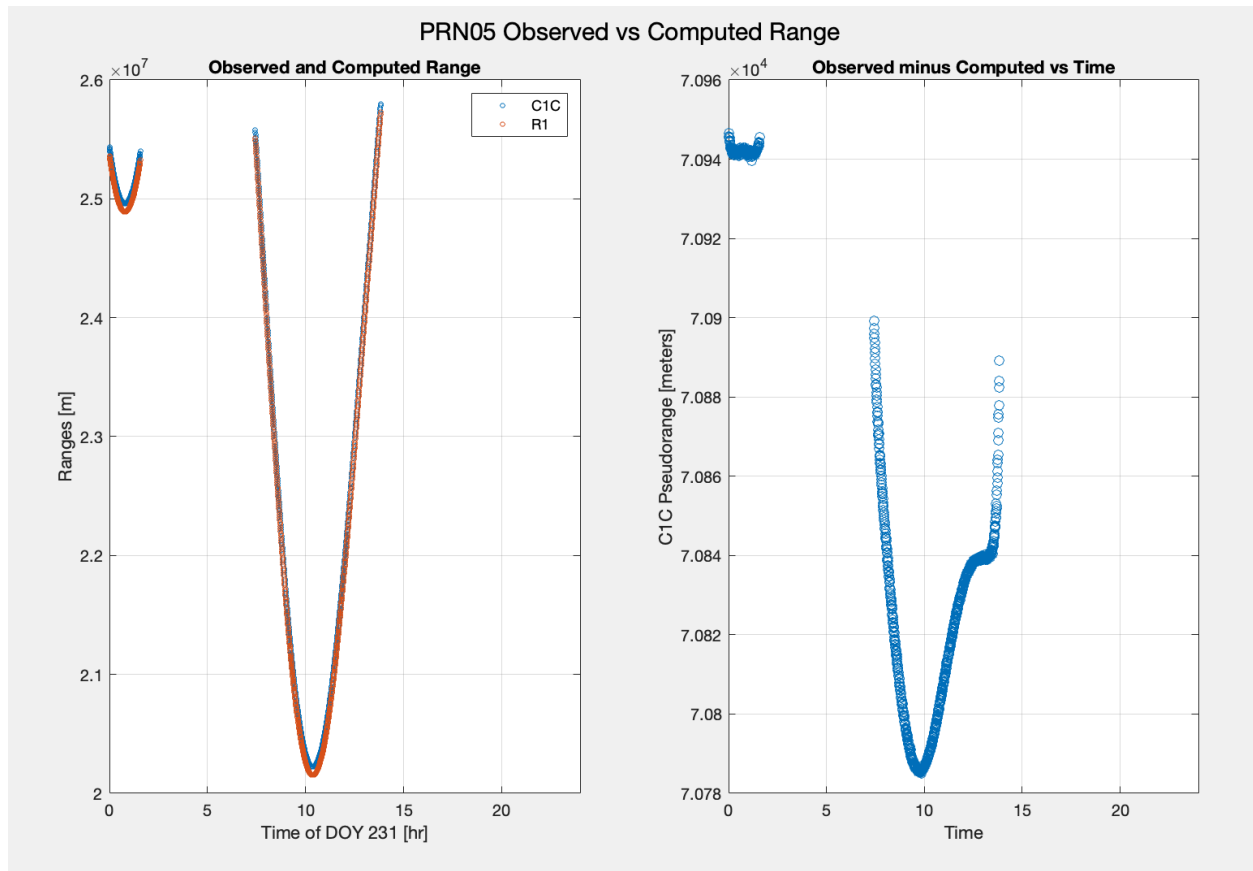
Now, taking that R0 range measurement and recomputing it based on travel times and earth's rotation, R1 is plotted. Additionally, R1-R0 is plotted on the right:



The largest difference between R1 and R0 is 78.9 meters.

Problem 4a

Here's a plot of observed vs computed (expected) range:



Problem 4b

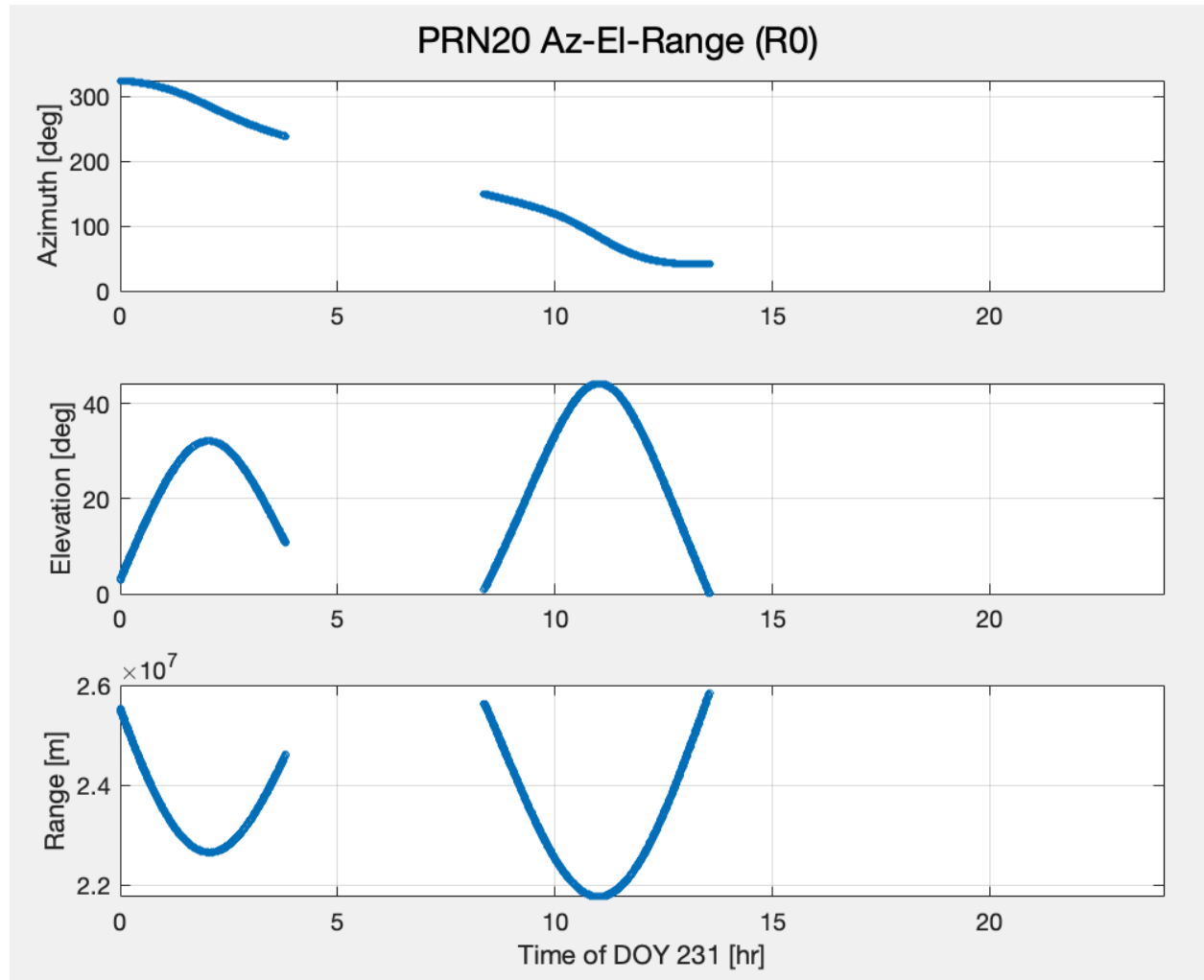
The largest source of difference between observed and computed values can be attributed to ionospheric and tropospheric effects and clock errors.

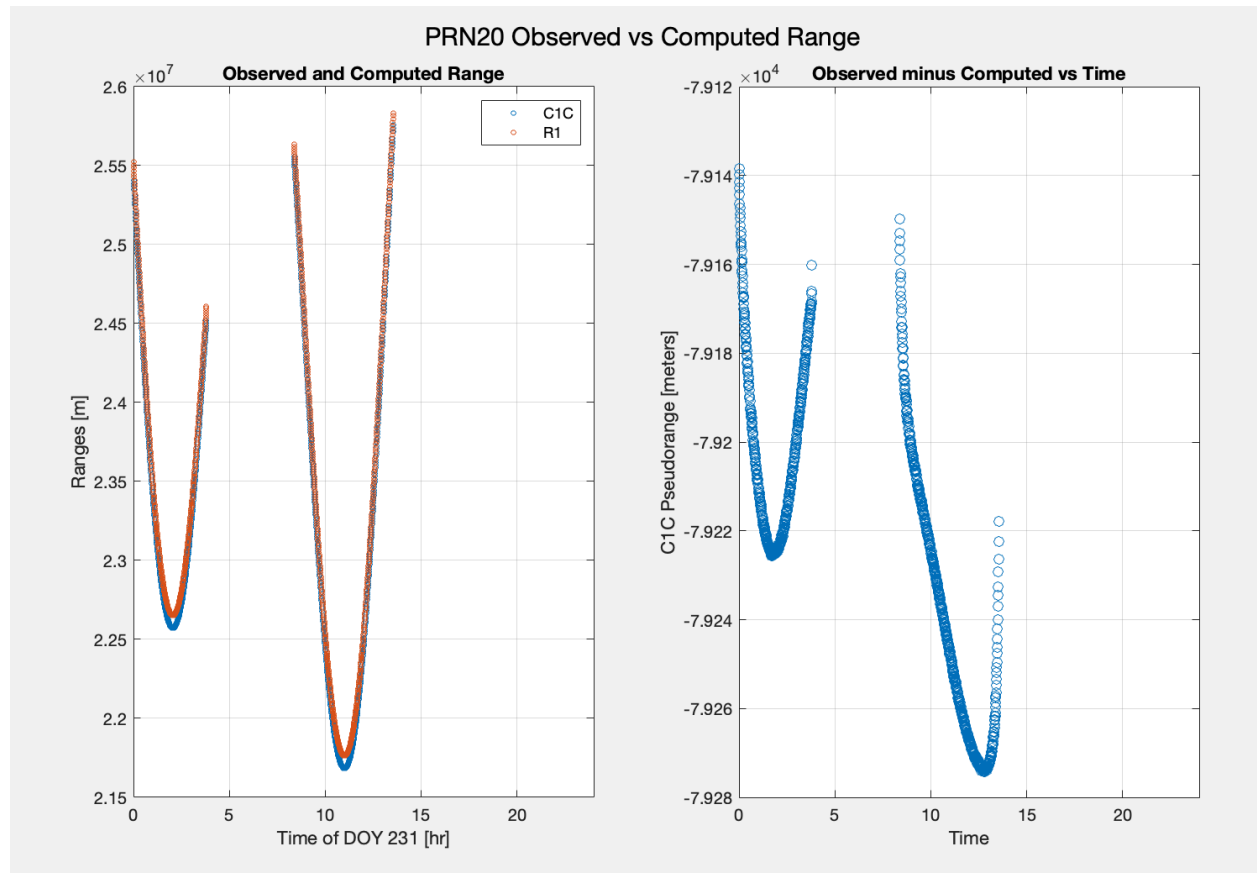
This can be seen in the fact that the difference is highest when the elevation angle is lowest. Since, low elevation means that the signal travels through more atmosphere, the atmospheric interference increases and so does the error.

This can be justified by looking at other satellites and seeing the same behavior. For example, for PRN20, similar behavior is seen (see Problem 5).

Problem 5

Picking PRN20 because that is from GPS Block III and visible during different times than PRN05.





As expected, the observed vs computed difference is highest when the satellite is near the horizon.

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```
clear; clc; close all;
```

ASEN 5090 - HW 3

Jash Bhalavat 09/20/2026

```
% Constants

% Speed of light in a vacuum
c = 2.99792458e8; % m/s, ME Table 4.1
```

Add path

```
addpath("HW3_Code/")
addpath("HW3_DATA/")
```

Problem 1a

```
% Set filename and use rinexread to extract GPS data
nist_rinex_filename = 'NIST00USA_R_20262310000_01D_30S_MO.rnx';
rinex_data = rinexread(nist_rinex_filename).GPS;

% Get time and convert data to arrays
time = rinex_data.Time;
rinex_data_array = table2array(rinex_data);

% Only extract PRN05 data and time
prn05_data = rinex_data_array(rinex_data_array(:,1) == 5, :);
prn05_time = time(rinex_data_array(:,1) == 5);
```

Problem 1b

```
figure(1)
plot(prn05_time, prn05_data(:,6), 'o', 'LineWidth',0.25)
xlabel("Time")
ylabel("C1C Pseudorange [meters]")
grid on
title("PRN05 C1C pseudorange vs Time")
```

Problem 1c

```
figure(2)
plot(prn05_time, prn05_data(:,13), 'o', 'LineWidth',0.25)
xlabel("Time")
ylabel("S1C Signal to Noise Ratio [dB-Hz]")
grid on
title("PRN05 S1C Signal to Noise Ratio vs Time")
```

Problem 1d

```
prn05_l1c_cycles = prn05_data(:,8); % [cycles]
l1_frequency = 1575.42e6; % [hz]
l1_wavelength = c / l1_frequency; % [meters]
prn05_l1c_meters = prn05_l1c_cycles * l1_wavelength;

figure(3)
subplot(2, 2, [1,3])
plot(prn05_time, prn05_data(:,6), 'o', 'LineWidth',0.25)
hold on
plot(prn05_time, prn05_data(:,28), 'o', 'LineWidth',0.25)
plot(prn05_time, prn05_l1c_meters, 'o', 'LineWidth',0.25)
legend("C1C", "C2L", "L1C")
xlabel("Time")
ylabel("Pseudorange [m]")
grid on
title("C1C, C2L, L1C (converted to meters) vs Time")

subplot(2,2,2)
plot(prn05_time, prn05_data(:,6)-prn05_data(:,28))
grid on
title("C1C minus C2L vs Time")
ylabel("Pseudorange [m]")
xlabel("Time")

subplot(2,2,4)
plot(prn05_time, prn05_data(:,6)-prn05_l1c_meters)
grid on
title("C1C minus L1C (converted to meters) vs Time")
ylabel("Pseudorange [m]")
xlabel("Time")
sgtitle("PRN05")
```

Problem 2a

```
% Read broadcast file and save data into array
broadcast_ephem_filename = 'brdc2310.26n';
ephem_data = read_clean_GPSbroadcast(broadcast_ephem_filename);
```

Problem 2b

```
filename = 'IGS00PSFIN_20262310000_01D_15M_ORB.SP3';
sp3_data = read_sp3(filename);

prn05_sp3 = sp3_data(sp3_data(:, 3) == 5, :); % [km]
time = (prn05_sp3(:,2) - prn05_sp3(1,2))./3600; % [hr]

week_number = 2432 * ones([length(time), 1]);
t_input = [week_number, prn05_sp3(:,2)];

[~, prn05_broadcast, ~, prn05_clock_bias] = eph2pvt2025(ephem_data, t_input, 5);

figure(1)
subplot(3,2,1)
plot(time, prn05_sp3(:,4)*1000, 'LineWidth',2)
hold on
plot(time, prn05_broadcast(:,1), 'LineWidth',2)
grid on
ylabel("X [m]")
xlim([0, 24])
legend("SP3", "Broadcast Ephemeris")
title("SP3 and Broadcast Ephemeris Positions")

subplot(3,2,3)
plot(time, prn05_sp3(:,5)*1000, 'LineWidth',2)
hold on
plot(time, prn05_broadcast(:,2), 'LineWidth',2)
grid on
xlim([0, 24])
ylabel("Y [m]")

subplot(3,2,5)
plot(time, prn05_sp3(:,6)*1000, 'LineWidth',2)
hold on
plot(time, prn05_broadcast(:,3), 'LineWidth',2)
grid on
ylabel("Z [m]")
xlim([0, 24])
xlabel("Time of DOY 231 [hr]")
sgtitle("PRN05 Position from Broadcast Ephemeris vs SP3")

subplot(3,2,2)
plot(time, prn05_sp3(:,4)*1000 - prn05_broadcast(:,1), 'LineWidth',2)
grid on
ylabel("X [m]")
xlim([0, 24])
title("SP3 minus Broadcast Ephemeris Positions")

subplot(3,2,4)
plot(time, prn05_sp3(:,5)*1000 - prn05_broadcast(:,2), 'LineWidth',2)
grid on
ylabel("Y [m]")
xlim([0, 24])

subplot(3,2,6)
plot(time, prn05_sp3(:,6)*1000 - prn05_broadcast(:,3), 'LineWidth',2)
grid on
ylabel("Z [m]")
xlabel("Time of DOY 231 [hr])"
```

Problem 2c

```
figure(2)
plot(time, prn05_clock_bias, 'LineWidth',2)
grid on
ylabel("Satellite Clock Bias [m]")
xlim([0, 24])
xlabel("Time of DOY 231 [hr]")
title("PRN05 Clock Bias from Broadcast Ephemeris")
```

Problem 3a

```
% Read broadcast file and save data into array
broadcast_ephem_filename = 'brdc2310.26n';
ephem_data = read_clean_GPSbroadcast(broadcast_ephem_filename);

% Set filename and use rinexread to extract GPS data
nist_rinex_filename = 'NIST00USA_R_20262310000_01D_30S_MO.rnx';
rinex_data = rinexread(nist_rinex_filename).GPS;

% Get time and convert data to arrays
time = rinex_data.Time;
rinex_data_array = table2array(rinex_data);

% Only extract PRN05 data and time
prn05_obs_data = rinex_data_array(rinex_data_array(:,1) == 5, :);
prn05_obs_time = time(rinex_data_array(:,1) == 5);
prn05_obs_time.TimeZone = 'UTC';

% Define the absolute GPS Epoch (January 6, 1980 is a Sunday)
gpsEpoch = datetime(1980, 1, 6, 'TimeZone', 'UTC');
gpsEpochArray = repmat(gpsEpoch, size(prn05_obs_time));

% Calculate total days elapsed since the GPS epoch
totalDays = floor(days(prn05_obs_time - gpsEpochArray));

% Find the GPS Week Number
gpsWeek = floor(totalDays / 7);

% Find the start of the current GPS week (the previous Sunday at 00:00:00)
currentSunday = gpsEpoch + days(gpsWeek * 7);

% Calculate Time of Week (TOW) in seconds
TOW = seconds(prn05_obs_time - currentSunday);

week_number = 2432 * ones([length(TOW), 1]);
t_input = [week_number, TOW];

[~, prn05_broadcast] = eph2pvt2025(ephem_data, t_input, 5);

[prn05_az_R0, prn05_el_R0, prn05_range_R0] = compute_azelrange(nist_ecef, prn05_broadcast);

plot_time = (t_input(:,2)-t_input(1,2))/3600;

figure(1)
subplot(3,1,1)
plot(plot_time, prn05_az_R0, 'o', 'MarkerSize',3)
grid on
ylabel("Azimuth [deg]")
xlim([0, 24])
```

```

subplot(3,1,2)
plot(plot_time, prn05_el_R0,'o','MarkerSize',3)
grid on
xlim([0, 24])
ylabel("Elevation [deg]")

subplot(3,1,3)
plot(plot_time, prn05_range_R0,'o','MarkerSize',3)
grid on
ylabel("Range [m]")
xlim([0, 24])
xlabel("Time of DOY 231 [hr]")
sgtitle("PRN05 Az-El-Range (R0)")

```

Problem 3b

```

prn05_range_R1 = compute_expected_range(t_input, prn05_broadcast, nist_ecef, c, ephemer_data, 5, omega_dot);

figure(2)
subplot(1,2,1)
plot(plot_time, prn05_range_R0,'o','MarkerSize',3)
hold on
plot(plot_time, prn05_range_R1,'o','MarkerSize',3)
grid on
ylabel("Range [m]")
xlim([0, 24])
xlabel("Time of DOY 231 [hr]")
legend("R0", "R1")
title("R0 and R1 ranges")

subplot(1,2,2)
plot(plot_time, prn05_range_R1 - prn05_range_R0,'o','MarkerSize',3)
grid on
ylabel("Range difference [m]")
xlim([0, 24])
xlabel("Time of DOY 231 [hr]")
title("R1 minus R0")

largest_difference = max(abs(prn05_range_R1 - prn05_range_R0));

```

Problem 4a

```

figure(3)
subplot(1,2,1)
plot(plot_time, prn05_obs_data(:,6), 'o', 'MarkerSize', 3)
hold on
plot(plot_time, prn05_range_R1, 'o', 'MarkerSize', 3)
grid on
xlim([0, 24])
ylabel("Ranges [m]")
xlabel("Time of DOY 231 [hr]")
legend("C1C", "R1")
title("Observed and Computed Range")
sgtitle("PRN05 Observed vs Computed Range")

subplot(1,2,2)
plot(plot_time, prn05_obs_data(:,6) - prn05_range_R1, 'o', 'LineWidth',0.25)
xlabel("Time")

```

```

xlim([0, 24])
ylabel("ClC Pseudorange [meters]")
grid on
title("Observed minus Computed vs Time")

```

Problem 5

```

% Read broadcast file and save data into array
broadcast_ephem_filename = 'brdc2310.26n';
ephem_data = read_clean_GPSbroadcast(broadcast_ephem_filename);

% Set filename and use rinexread to extract GPS data
nist_rinex_filename = 'NIST00USA_R_20262310000_01D_30S_MO.rnx';
rinex_data = rinexread(nist_rinex_filename).GPS;

% Get time and convert data to arrays
time = rinex_data.Time;
rinex_data_array = table2array(rinex_data);

% Only extract PRN20 data and time
prn20_obs_data = rinex_data_array(rinex_data_array(:,1) == 20, :);
prn20_obs_time = time(rinex_data_array(:,1) == 20);
prn20_obs_time.TimeZone = 'UTC';

% Define the absolute GPS Epoch (January 6, 1980 is a Sunday)
gpsEpoch = datetime(1980, 1, 6, 'TimeZone', 'UTC');
gpsEpochArray = repmat(gpsEpoch, size(prn20_obs_time));

% Calculate total days elapsed since the GPS epoch
totalDays = floor(days(prn20_obs_time - gpsEpochArray));

% Find the GPS Week Number
gpsWeek = floor(totalDays / 7);

% Find the start of the current GPS week (the previous Sunday at 00:00:00)
currentSunday = gpsEpoch + days(gpsWeek * 7);

% Calculate Time of Week (TOW) in seconds
TOW = seconds(prn20_obs_time - currentSunday);

week_number = 2432 * ones([length(TOW), 1]);
t_input = [week_number, TOW];

[~, prn20_broadcast] = eph2pvt2025(ephem_data, t_input, 20);

[prn20_az_R0, prn20_el_R0, prn20_range_R0] = compute_azelrange(nist_ecef, prn20_broadcast);

plot_time = (t_input(:,2)-t_input(1,2))/3600;

figure(1)
subplot(3,1,1)
plot(plot_time, prn20_az_R0, 'o', 'MarkerSize', 3)
grid on
ylabel("Azimuth [deg]")
xlim([0, 24])

subplot(3,1,2)
plot(plot_time, prn20_el_R0, 'o', 'MarkerSize', 3)
grid on
xlim([0, 24])

```

```

ylabel("Elevation [deg]")

subplot(3,1,3)
plot(plot_time, prn20_range_R0,'o','MarkerSize',3)
grid on
ylabel("Range [m]")
xlim([0, 24])
xlabel("Time of DOY 231 [hr]")
sgtitle("PRN20 Az-El-Range (R0)")

prn20_range_R1 = compute_expected_range(t_input, prn20_broadcast, nist_ecef, c, ephemer_data, 20, omega_dot);

figure(2)
subplot(1,2,1)
plot(plot_time, prn20_range_R0,'o','MarkerSize',3)
hold on
plot(plot_time, prn20_range_R1,'o','MarkerSize',3)
grid on
ylabel("Range [m]")
xlim([0, 24])
xlabel("Time of DOY 231 [hr]")
legend("R0", "R1")
title("R0 and R1 ranges")

subplot(1,2,2)
plot(plot_time, prn20_range_R1 - prn20_range_R0,'o','MarkerSize',3)
grid on
ylabel("Range difference [m]")
xlim([0, 24])
xlabel("Time of DOY 231 [hr]")
title("R1 minus R0")

largest_difference = max(abs(prn20_range_R1 - prn20_range_R0));

figure(3)
subplot(1,2,1)
plot(plot_time, prn20_obs_data(:,6), 'o', 'MarkerSize', 3)
hold on
plot(plot_time, prn20_range_R1, 'o', 'MarkerSize', 3)
grid on
xlim([0, 24])
ylabel("Ranges [m]")
xlabel("Time of DOY 231 [hr]")
legend("C1C", "R1")
title("Observed and Computed Range")
sgtitle("PRN20 Observed vs Computed Range")

subplot(1,2,2)
plot(plot_time, prn20_obs_data(:,6) - prn20_range_R1, 'o', 'LineWidth',0.25)
xlabel("Time")
xlim([0, 24])
ylabel("C1C Pseudorange [meters]")
grid on
title("Observed minus Computed vs Time")

```

Remove path

```

rmpath("HW2_Code/")
rmpath("HW2_DATA/")
rmpath("HW3_Code/")

```

```
rmpath("HW3_DATA/")
rmpath("../HW 2")
```

FUNCTIONS

```
function R1 = compute_expected_range(t_r, r_gps_tr, r_rx, c, ephem_data, prn, omega_E)
    % Computed expected range given Tr [WN, TOW], r_gps [ECEF, m], c [m/s]
    max_count = 5;

    % Initial measure of time taken for signal to get from satellite to
    % receiver
    dt = norm(r_gps_tr(1,:) - r_rx)/c;

    for j = 1:max_count
        for i = 1:length(t_r)
            % Step 2
            R(i) = norm(r_gps_tr(i,:) - r_rx);

            % Step 3
            t_t(i,1) = t_r(i,2) - R(i)/c;
        end
        t_t = [t_r(:,1), t_t];

        % Step 4
        [~, r_gps_tt] = eph2pvt2025(ephem_data, t_t, prn);

        % Step 5
        for i = 1:length(t_r)
            phi = omega_E * (t_r(i,2)-t_t(i,2));
            r_gps_tr(i,:) = ([cos(phi), sin(phi), 0; -sin(phi), cos(phi), 0; 0, 0, 1] * r_gps_tt(i,:))';

            % Step 6
            R1(i,1) = norm(r_gps_tr(i,:) - r_rx);
        end

        % Calculate new signal travel time
        dt_new = R1(1)/c;

        % Compare signal travel times and break if within tolerance
        if abs(dt_new - dt) < 1e-12
            break
        end

        % If not within tolerance, update current dt
        dt = dt_new;
    end
end
```